

The Irrationality of $\sqrt{2}$

Question

Prove that $\sqrt{2}$ is irrational. In other words, prove that there do not exist integers p and q , with $q \neq 0$, such that

$$\sqrt{2} = \frac{p}{q}.$$

Equivalently, prove that 2 is not a perfect square in $\mathbb{Q}$, or that the polynomial $x^2 - 2$ has no rational root.

Solution

Theorem. *The real number $\sqrt{2}$ is irrational. Equivalently, no integers p, q with $q \neq 0$ satisfy $\sqrt{2} = p/q$.*

Proof. Suppose, for contradiction, that $\sqrt{2}$ is rational. Then there exist integers a, b with $b \neq 0$ such that

$$\sqrt{2} = \frac{a}{b}.$$

Replacing both a and b by their negatives if necessary, we may assume that $b > 0$. Let

$$d = \gcd(|a|, b).$$

Since d divides both a and b , the numbers

$$p = \frac{a}{d} \quad \text{and} \quad q = \frac{b}{d}$$

are integers. Moreover, $q > 0$ and

$$\sqrt{2} = \frac{p}{q}.$$

This representation is in lowest terms. Indeed, if a positive integer c divided both p and q , then cd would divide both $a = dp$ and $b = dq$. Since d is their greatest positive common divisor, we would have $cd \leq d$, and hence $c \leq 1$. Thus $c = 1$, so

$$\gcd(|p|, q) = 1.$$

We first prove the parity fact that if the square of an integer is even, then the integer itself is even. Let $n \in \mathbb{Z}$. If n were odd, then $n = 2k + 1$ for some $k \in \mathbb{Z}$, and therefore

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1,$$

which is odd. Consequently, an integer whose square is even cannot be odd and must therefore be even.

Squaring the equality $\sqrt{2} = p/q$ gives

$$2 = \frac{p^2}{q^2}.$$

Because $q \neq 0$, we may multiply by q^2 to obtain

$$p^2 = 2q^2.$$

Thus p^2 is even. By the parity fact just proved, p is even, so $p = 2r$ for some $r \in \mathbb{Z}$. Substituting this into $p^2 = 2q^2$ yields

$$(2r)^2 = 2q^2,$$

and hence

$$4r^2 = 2q^2.$$

Dividing by 2 gives

$$q^2 = 2r^2.$$

Therefore q^2 is even, so the same parity fact shows that q is even.

Hence 2 divides both p and q . Thus 2 is a positive common divisor of $|p|$ and q , which implies

$$\gcd(|p|, q) \geq 2.$$

This contradicts the lowest-terms condition $\gcd(|p|, q) = 1$. Therefore no integers p, q with $q \neq 0$ satisfy $\sqrt{2} = p/q$, and consequently $\sqrt{2}$ is irrational. $\square$